SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	CSE	**Date:**	10/10/2026

Code of Conduct

I N.Sai Sankar(192311187) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: N.Sai sankar

Instructions

- Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
- Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q1**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**

9. Uninformed Search Strategies	Error tracing of BFS, DFS and uniform-cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Annexure A – Error Analysis Task

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below on the foundations of AI and intelligent agents. The notes contain errors; some statements may be correct. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the "thinking rationally" approach to AI.
S2	The "AI winters", periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term "artificial intelligence".
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Identify which statements are correct and which are incorrect. For each incorrect statement, quote the erroneous phrase or claim (one statement contains more than one error).

Statement	Correct / Incorrect	Erroneous phrase / claim
S1		
S2		
S3		
S4		
S5		
S6		

(b) Root cause analysis: for every error, explain the underlying misconception, i.e. which concept has been misunderstood and why the statement goes wrong.

Error (statement / part)	Root cause (misconception)

(c) Rewrite each faulty statement correctly and justify your correction. For a statement that is already correct, write "No change" and justify why.

Statement	Corrected statement	Justification
S1		
S2		
S3		
S4		
S5		
S6		

(d) Apply the corrected concepts to the following case. A self-driving taxi chooses a highway route that every available traffic report showed to be clear; an unforeseeable landslide then blocks the road.

(i) Using the four factors that decide whether an agent is rational, argue whether the taxi's choice was rational.

(ii) Name the properties of the taxi's task environment that make perfect foresight impossible.

(iii) Describe one situation in which a simple reflex taxi would behave poorly, and state what a model-based reflex taxi keeps in order to avoid it.

(i)	
(ii)	

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

A student formulates the **Missionaries and Cannibals** problem for a problem-solving agent. Three missionaries and three cannibals stand on the left bank of a river, with a boat that carries one or two people and cannot cross empty. The boat is on the left bank at the start. Missionaries must never be outnumbered by cannibals on either bank (a bank with no missionaries is safe). The objective is to move everyone to the right bank using the **fewest crossings**.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	M = C
Legality check	A state is legal if C ≤ M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search → goal formulation → problem formulation → execution.
Note N2	The state space contains 16 states, of which 10 are legal.

(a) For each component and note, state whether it is correct or incorrect; for each incorrect one, state what is wrong.

Component / Note	Correct / Incorrect	What is wrong
State representation		
Initial state		
Actions		
Goal test		
Legality check		
Path cost		
Note N1		
Note N2		

(b) Root cause analysis: for every error, explain the underlying cause, i.e. which principle of problem formulation or of the problem-solving agent has been violated.

Error (component / note)	Root cause

(c) Give the corrected version of every faulty component or note and justify each correction.

Component / Note	Corrected formulation	Justification

(d) Apply your corrected formulation.

(i) Complete the table of boat loads available from the initial state, writing each resulting state as (M, C, boat side) and whether it is legal, with the reason.

Boat load	Resulting state (M, C, boat)	Legal? Reason
1M		
2M		
1C		
2C		
1M + 1C		

(ii) Give one state that the student's legality check accepts although it is actually illegal.

(iii) Using your corrected path cost, state the cost of an optimal solution.

(ii)	
(iii)	

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

A student ran BFS, DFS and uniform-cost search (UCS) on the weighted undirected graph below. The start node is S and the goal node is G. **Conventions:** graph search with an explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties broken alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them. **BFS:** do not add a node that is already explored or already in the queue. **DFS:** use a stack with its top on the left; push children so that the alphabetically first child is popped first; do not push a node that is already explored or already on the stack. **UCS:** write entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found.

Edge	Cost	Edge	Cost
S – C	3	B – G	3
S – D	3	D – E	2
A – B	2	D – G	6
A – C	6	E – G	3
B – D	4		

Student's BFS trace (queue initially [S])

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A]	{S, C, D, B}
5	E	[G, A]	{S, C, D, B, E}
6	G (goal)	[A]	{S, C, D, B, E}

BFS result: path S – D – G, cost 9, 6 nodes removed from the frontier.

Student's DFS trace (stack initially [S], top on the left)

Iteration	Node Popped	Stack after Pushing Children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S – D – G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	Node Expanded	g(n)	Priority Queue after Expansion

1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)
5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	–

UCS result: path S – D – G, cost 9, 7 nodes removed from the frontier.

Student's remarks

No.	Remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S – D – G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

(a) Examine the three traces and the remarks. List every error, giving the trace and iteration (or the remark number) and what is wrong. An error that is only carried forward counts once.

No.	Where (trace / iteration / remark)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the rule or concept that was violated and how it led to the wrong entry or conclusion.

No.	Root cause (rule or concept violated)
1	
2	
3	

4	
5	
6	
7	

(c) Write the corrected traces and justify each correction.

(i) BFS: the corrected queue after iteration 3 and the full order in which nodes are dequeued.

Queue after iteration 3	
Order of dequeued nodes	

(ii) DFS: the corrected trace.

Iteration	Node Popped	Stack after Pushing Children
1		
2		
3		
4		
5		

(iii) UCS: the corrected trace.

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1			
2			
3			
4			
5			
6			

Justification of the corrections	

(d) Apply the corrected traces.

(i) Complete the comparison table.

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS				
DFS				
UCS				

(ii) Use iteration 3 of the corrected BFS trace to explain why an explored set is needed in graph search.

(ii)	

Question 4: Search with Partial Information – Error Analysis of Belief-State Computation

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search)

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). The goal is that both squares are Clean. Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

Task A – sensorless agent. The agent knows it is in the Left square but nothing about the dirt, so its initial belief state is {S1, S3, S5, S7}. A student applied the plan [Suck, Right, Suck]:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Task B – agent with a local sensor that reports [location, status of the current square]. Starting from the belief state of all eight states, the agent receives the percept [Left, Dirty], performs Suck, receives [Left, Clean], performs Right and receives [Right, Dirty]. The student's table:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

Student's conclusions

No.	Conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.

C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

(a) Examine the tables and the conclusions. List every error, giving the task and step (or the conclusion number) and what is wrong. An error that is only carried forward from an earlier wrong row counts once.

No.	Where (task / step / conclusion)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the concept or rule of belief-state search that was misapplied.

No.	Root cause (concept or rule misapplied)
1	
2	
3	
4	
5	
6	
7	

(c) Write the corrected tables and conclusions, with justification.

(i) Task A: the corrected belief states.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	

2	Right	
3	Suck	

(ii) Task B: the corrected belief states after each action and each percept.

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	
1	Suck		[Left, Clean]	
2	Right		[Right, Dirty]	

(iii) Corrected conclusions C1 and C2, with justification.

C1	
C2	

(d) Apply the corrected concepts.

(i) How many different belief states are possible when there are 8 physical states? Why does this make belief-state search harder than search over physical states?

(i)	

(ii) For the initial belief state {S1, S3, S5, S7}, apply the plan [Right, Suck], write the belief state after each action, and state whether it is a solution.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Right	
2	Suck	
	Solution? Why?	

ANSWERS

Question 1:

(a) Identification

Statement	Correct / Incorrect	Erroneous phrase / claim
S1	Incorrect	"internal reasoning always follows the laws of logic" and "standard example of the thinking rationally approach" (two errors)
S2	Incorrect	"AI winters… occurred in the 1950s, before 1956" and "workshop then revived the field"
S3	Incorrect	"always achieves the best possible outcome because it knows the actual result in advance" and "was not rational"
S4	Incorrect	Performance measure and Actuators are swapped
S5	Incorrect	"partially observable, stochastic, episodic" (static and discrete are correct)
S6	Correct	None

(b) Root cause analysis

Error (statement / part)	Root cause (misconception)
S1	The Turing Test is a behavioural test (can the machine's answers be mistaken for a human's). It says nothing about internal reasoning.
S2	The AI winters came **after** Dartmouth (mid-1970s and late 1980s to early 1990s), when over-optimism went unmet. Dartmouth (1956) **launched** the field and named it, so there was nothing to revive.
S3	Confuses rationality with omniscience. A rational agent maximizes **expected** performance given what it perceives and knows, not the actual outcome.
S4	Confuses what is *measured* (performance measure) with what the agent *acts through* (actuators).
S5	Chess without a clock is fully observable (whole board visible), deterministic (no chance), and sequential (each move affects later ones).

(c) Corrected statements

Statement	Corrected statement	Justification
S1	The Turing Test (1950) judges a machine intelligent if a human interrogator cannot tell its responses from a human's. It is the standard example of the **acting humanly** approach.	It tests external behaviour only, not reasoning.
S2	The AI winters occurred **after** the 1956 Dartmouth workshop (1970s and late 1980s to early 1990s). The workshop launched AI as a field and coined the term.	Dartmouth is the starting point, so winters could only follow it.
S3	A rational agent selects the action expected to maximize its performance measure given its percept sequence and built-in knowledge. It is not omniscient, so the taxi **was rational**.	Rationality concerns expected outcomes. An unforeseeable event does not make a sensible decision irrational.
S4	**Performance measure:** safe, fast, legal, comfortable trip. **Environment:** roads, traffic, pedestrians, weather. **Actuators:** steering, accelerator, brake, horn. **Sensors:** cameras, GPS, speedometer.	The performance measure defines success. Actuators are the output devices.
S5	Chess without a clock is **fully observable, deterministic, sequential, static and discrete** (multi-agent, competitive).	The board is fully visible, moves have no randomness, and moves have long-term consequences.
S6	No change.	This is the correct definition of a model-based reflex agent. The internal state carries information the current percept lacks.

(d) Application

(i) Was the taxi rational? The four factors:

- **Performance measure:** a safe, fast trip. The highway looked fastest and safe.
- **Prior knowledge:** the road network and how traffic usually behaves.
- **Available actions:** the routes it could choose.
- **Percept sequence:** every traffic report said the road was clear.

Given these four factors, choosing the highway maximized expected performance, so the choice was **rational**. The landslide could not have been predicted, and rationality is not omniscience.

(ii) Properties preventing perfect foresight: the environment is **partially observable** (hazards ahead can't be seen), **stochastic** (events are uncertain) and **dynamic** (the road changes while driving). It is also continuous and multi-agent.

(iii) Simple reflex vs model-based: a simple reflex taxi reacts only to the current percept. For example, it changes lanes when its camera shows the adjacent lane empty. It forgets a car it saw in the mirror a moment ago, which is now in its blind spot, so it can cut that car off. A model-based reflex taxi keeps an **internal state**: positions and speeds of nearby vehicles from earlier percepts, plus the effects of its own actions. That state lets it infer the hidden car.

Question 2:

(a) Correct or incorrect

Component / Note	Correct / Incorrect	What is wrong
State representation	Incorrect	The boat's side is missing. The same (M, C) can occur with the boat on either bank.
Initial state	Incorrect	(3,3) omits the boat. It should be (3,3,L). Carried forward from the state representation.
Actions	Correct	The five loads are complete. Applicability depends on the boat side (see corrections).
Goal test	Incorrect	M = C is true in many states such as (2,2) and (1,1). The goal needs everyone on the right bank.
Legality check	Incorrect	It checks only the left bank. The right bank must also be safe.
Path cost	Incorrect	The objective is the fewest crossings, so cost is 1 per crossing, not people carried.
Note N1	Incorrect	Order is wrong. Goal formulation must come first, then problem formulation, then search, then execution.
Note N2	Incorrect	The count ignores the boat. The true space is $4 \times 4 \times 2 = 32$ states, of which 20 are legal.

(b) Root cause analysis

Error (component / note)	Root cause
State representation / initial state	A state must contain all information needed to determine legal actions and the goal. The boat position is part of that.
Goal test	The goal test must be true only for goal states. M = C does not describe "all on the right bank".
Legality check	The constraint applies to **both** banks. Checking one bank ignores half the constraint.
Path cost	The cost function must match the stated objective (fewest crossings). Counting people optimizes something else.
N1	A problem-solving agent first formulates a goal, then the problem, then searches. A problem cannot be formulated before a goal exists, and you cannot search before the problem is defined.
N2	The state space was counted without the boat. Counting (M, C) alone gives 16 combinations, which is not the true state space.

(c) Corrected formulation

Component / Note	Corrected formulation	Justification
State representation	(M, C, B): missionaries and cannibals on the left bank, B ∈ {L, R}	The boat position determines which bank people can leave from.
Initial state	(3, 3, L)	Everyone and the boat start on the left.
Actions	Move 1M, 2M, 1C, 2C or 1M+1C from the boat's bank to the other bank. The load must be available on that bank.	Applicability depends on B.
Goal test	State = (0, 0, R)	This is the only state with everyone on the right.
Legality	Left: M = 0 or C ≤ M. Right (3−M, 3−C): 3−M = 0 or 3−C ≤ 3−M.	Both banks must be safe.
Path cost	1 per crossing	This minimizes the number of crossings.
N1	Goal formulation → problem formulation → search → execution	Standard agent cycle.
N2	4 × 4 × 2 = **32** states, **20** legal (10 safe (M, C) pairs × 2 boat sides)	Safe pairs: (0,0), (0,1), (0,2), (0,3), (3,0), (3,1), (3,2), (3,3), (1,1), (2,2).

(d) Application

(i) Boat loads from (3,3,L):

Boat load	Resulting state (M, C, boat)	Legal? Reason
1M	(2,3,R)	No. Left has 2M and 3C, so cannibals outnumber missionaries.
2M	(1,3,R)	No. Left has 1M and 3C.
1C	(3,2,R)	Yes. Left is 3M vs 2C; right is 0M with 1C (safe).
2C	(3,1,R)	Yes. Left is 3M vs 1C; right is 0M with 2C (safe).
1M + 1C	(2,2,R)	Yes. Left is 2 vs 2; right is 1 vs 1.

(ii) State (2,1,·). The student's check passes (left has 1C ≤ 2M), but the right bank holds 1M and 2C, so the missionary is outnumbered. (1,0,·) also works: the right bank would have 2M and 3C.

(iii) The optimal solution has **11 crossings**, so the cost is **11**.

Question 3: Searching for Solutions and Uninformed Search

Adjacency (alphabetical): S: C, D | A: B, C | B: A, D, G | C: A, S | D: B, E, G, S | E: D, G | G: B, D, E

Question 3:

(a) Errors

No.	Where	What is wrong
1	BFS iteration 3	Queue is [B, E, G, A]. A was already enqueued in iteration 2, so FIFO order gives [A, B, E, G]. Later rows are carried forward.
2	DFS iteration 1	Stack is [D, C]. C must be on top, so it should be [C, D].
3	DFS iteration 2	The same push-order mistake is repeated. Children must be pushed with the alphabetically first on top.
4	UCS iteration 4	Expanding E reaches G at 5 + 3 = 8, cheaper than G:9(D). The entry must be replaced by G:8(E). Iterations 5 to 7 are carried forward.
5	Remark R1	False. The graph is undirected, so BFS can reach already-dequeued nodes again.
6	Remark R2	The three strategies do not all return the same path, and they are not all optimal.
7	Remark R3	Fewest nodes removed is not a measure of the best strategy.

(b) Root cause analysis

No.	Root cause (rule or concept violated)
1	BFS uses a FIFO queue. New nodes join behind those already waiting, so A (already queued) stays ahead of B, E, G.
2	The DFS convention requires the alphabetically first child to be popped first. A stack pops the last item pushed, so the first child must be pushed last (on top).
3	The same stack-order rule was violated again. This is an independent repeat of error 2.
4	UCS must replace a frontier entry whenever a cheaper path is found. Skipping the update left a stale g = 9 for G and made the final path wrong.
5	Without an explored set, undirected edges send BFS back to parents (e.g. D → S). Graph search needs the explored set to avoid duplicates and loops.
6	BFS finds the shallowest goal, which is optimal only for unit step costs. DFS is not optimal. Only UCS is optimal for general costs. Matching costs coincided only because the traces were wrong.
7	Efficiency in nodes removed is not solution quality. DFS's cost and optimality must also be considered.

(c) Corrected traces

BFS

Item	Answer
Queue after iteration 3	[A, B, E, G]
Order of dequeued nodes	S, C, D, A, B, E, G

Iter	Dequeued	Queue after expansion	Explored set
1	S	[C, D]	{S}

Iter	Dequeued	Queue after expansion	Explored set
2	C	[D, A]	{S, C}
3	D	[A, B, E, G]	{S, C, D}
4	A	[B, E, G]	{S, C, D, A}
5	B	[E, G]	{S, C, D, A, B}
6	E	[G]	{S, C, D, A, B, E}
7	G (goal)	–	–

DFS

Iteration	Node popped	Stack after pushing children
1	S	[C, D]
2	C	[A, D]
3	A	[B, D]
4	B	[G, D]
5	G (goal)	[D]

UCS

Iteration	Node expanded	g(n)	Priority queue after expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), G:8(E), A:9(C)
5	B	7	G:8(E), A:9(C)
6	G (goal)	8	–

In UCS iteration 5, B reaches A at cost 9, which only ties the existing entry, and reaches G at cost 10, which is worse than 8. Neither entry changes.

Justification: BFS uses FIFO order. DFS pushes children so the alphabetically first is popped first. UCS replaces any frontier entry whenever a cheaper path is found.

(d) Application

(i) Comparison table

Strategy	Path found	Path cost	Nodes removed from frontier	Optimal here?
BFS	S–D–G	9	7	No
DFS	S–C–A–B–G	14	5	No
UCS	S–D–E–G	8	6	Yes

(ii) In iteration 3, D is dequeued and its neighbours are B, E, G and **S**. The edge S–D is undirected, so S is a neighbour of D. S is already explored, so it must not go back into the queue. Without the explored set, S would be re-added and the search would loop or duplicate work. The explored set is what blocks it.

Question 4:

(a) Errors

No.	Where	What is wrong
1	Task A, step 3	After Suck in the Right square, S6 → S8. The belief state is {S8}, not {S6, S8}.
2	Conclusion C1	The plan [Suck, Right, Suck] ends in {S8}, which is the goal. The agent also ends up knowing everything. Sensorless problems are solvable.
3	Task B, step 0 (updated)	Percept [Left, Dirty] leaves {S1, S3}. S2 and S6 have the agent on the right.
4	Task B, step 1 (predicted)	Carried forward from error 3. Correct value is {S5, S7}. Not counted separately.
5	Task B, step 2 (predicted)	S7 moves to S8 on Right, not S7. The correct result is {S6, S8}.
6	Conclusion C2	S6 = (R, C, D) has a dirty right square, so the goal has **not** been reached.
7	Conclusion C3	Belief states are subsets of physical states, so there are $2^8 = 256$ of them.

(b) Root cause analysis

No.	Root cause (concept or rule misapplied)
1	Suck cleans the **current** square. In S6 the agent is on the right, so the right square gets cleaned.
2	Coercion: a sensorless agent can reach a known state by acting on all possibilities at once. Success means every state in the final belief state is a goal state.
3	Percept update must keep only states consistent with the percept (agent location **and** local dirt status).
4	Carried forward from error 3.
5	Prediction applies the action to **every** state in the belief state. S7 has the agent on the left, so Right moves it to S8.
6	Knowing the state is different from being in a goal state. A goal belief state needs every member to satisfy the goal.
7	A belief state is a *set* of physical states, so the belief space is the power set, which is exponentially larger.

(c) Corrected tables and conclusions

Task A

Step	Action	Belief state after action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	**{S8}**

Task B

Step	Action	After action (predicted)	Percept	After percept (updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	**{S1, S3}**
1	Suck	**{S5, S7}**	[Left, Clean]	{S5, S7}
2	Right	**{S6, S8}**	[Right, Dirty]	{S6}

Conclusion	Corrected version with justification
C1	Task A: after the plan the belief state is the singleton {S8}, which is the goal, so the agent knows the state and has succeeded. Sensorless problems **can** be solved by a plan that works from every possible start state.
C2	Task B: the belief state is the singleton {S6}, so the agent knows its location and the dirt status, but the right square is still dirty, so the goal is **not** reached. A further Suck gives S8.

(d) Application

(i) There are 2^8 = **256** possible belief states (255 non-empty), compared with only 8 physical states. With N physical states there are 2^N belief states. The search space grows exponentially, and each node is a whole set to store and compare, which is why belief-state search is much harder.

(ii) Plan [Right, Suck] from {S1, S3, S5, S7}:

Step	Action	Belief state after action
0	(start)	{S1, S3, S5, S7}
1	Right	{S2, S4, S6, S8}
2	Suck	{S4, S8}
	Solution? Why?	**No.** S4 = (R, D, C) still has a dirty left square, so not every state in the final belief state is a goal state.

Rubrics for Error Analysis

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Error Identification	**20**	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	**30**	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	**25**	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	**15**	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	**10**	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____